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Chapter

5

Rasch-CDM: A Combination of Rasch and Cognitive Diagnosis Models to Assess a Learning Progression

Yizhu Gao, Xiaoming Zhai, Ahra Bae, Wenchao Ma

Abstract

Learning progressions (LPs) have been highlighted in science education over the past ten years. LPs have the potential to guide curriculum development, instruction, and assessments. An LP has to be validated using empirical evidence, and the Rasch model is one of the most broadly used measurement models to connect the empirical evidence with the hypothesized LP. However, the Rasch model has limitations as it only provides students' overall ability measures and item difficulties, offering limited information about the specific cognitive skills (i.e., attributes) needed to complete specific instruments. This study thus introduced a novel approach, Rasch-CDM, by integrating the Rasch model and cognitive diagnosis model (CDM) to assess students' ability and fine-grained attributes. Students' ability, the difficulty of individual attributes, as well as students' attribute mastery patterns collected from the Rasch-CDM approach can be used to validate LPs. Additionally, attribute mastery patterns of LP levels as well as ability distribution and attribute difficulty can be visualized on a map, named as MGZA, showing how students progress towards more sophisticated thinking about a topic. To demonstrate this approach, we provide an example study—validating an LP of buoyancy using the Rasch-CDM approach.

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1 Introduction

Learning progressions (LPs), which depict children’s “successively more sophisticated ways of thinking about a topic that can follow one another as children learn about and investigate the topic” (NRC, 2007, p. 219), have drawn increasing attention in science education in the past decade. Research on LPs primarily focuses on examining students’ cognitive development of thinking during scientific practices, aiming to provide evidence for meaningful curriculum development, instruction, and assessments. However, examining students’ LP faces challenges. One challenge is due to the nature of children’s cognitive development, which takes years and progresses diversely depending on the instruction (Gao et al., 2021). Such a continuous progression challenges researchers in regard to the efforts placed to track the progression, particularly if students’ progression trajectories are diverse. It would be ideal if researchers could continually observe students’ science performances, collect evidence, and infer students’ cognitive development in a considerable time span. However, such work would be time and cost-intensive, and exemplary longitudinal research does not yet exist (Jin et al., 2019).

Instead, many researchers in science education tend to rely on theoretical hypotheses and statistical approaches such as the Rasch model to assess students’ LP. For example, Neumann et al. (2013) applied the Rasch model to calibrate students’ understanding of energy and validated a four-level progression hypothesis. Researchers prefer Rasch because of many advantages it possesses, such as transforming non-interval raw scores into interval and linear student abilities (Boone, 2016; Liu, 2010). As Rasch modeling is broadly used in LP research, researchers realized that this approach overly relies on the overall construct of measurement instrument (Liu, 2020, p. 1), and may lack the power to detect the nuance of students’ diverse cognitive abilities required for completing the instrument. For example, the item below may require students to develop multiple cognitive skills (Gao et al., 2020) to respond to the item, such as (a) being able to identify buoyancy (and gravity), (b) knowing the direction of buoyancy (upward, which is opposite to gravity and normal force from the bottom of the container), and (c) knowing that objects stay at rest only if the net force is zero (Gao et al., 2020).

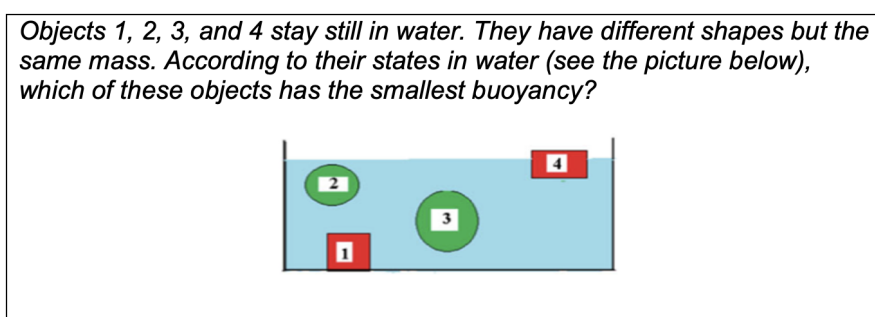


Figure 5.1. An example item

Rasch modeling could calibrate students’ overall ability based on their performance on a test including items like the example, but it is difficult to assess students’ specific cognitive skills required to accomplish an individual item. More importantly,

measurement evidence is needed for assessment design and development; otherwise, we might not be able to develop assessment items that assess cognitive skills as intended. It is possible that additional cognitive skills are also required but may be ignored by assessment designers. To overcome these challenges, this study introduces an approach (i.e., Rasch-CDM) incorporating the Rasch model and cognitive diagnosis model (CDM) that can assess fine-grained cognitive skills to validate students' LPs. CDM could examine the individual cognitive skills required for one assessment item (Zhai, Haudek, Ma, 2022), and thus could supplement the Rasch model that can calibrate students' overall ability based on performance on a test.

In our prior studies (e.g., Gao et al., 2020, 2021), we applied a similar framework in examining middle school students' LP of buoyancy, but the methodologies were not the focus of the publications. This posed challenges for science education researchers to fully understand the novel approach and the findings.

In this study, we introduce the Rasch-CDM approach for assessing LPs. We aim to provide science education researchers access to the Rasch-CDM approach using plain language that articulates its rationale and procedures. We also introduce the GDINA R package (Ma & de la Torre, 2019) that was developed for CDM. We present an example to demonstrate the approach and discuss the difference between the Rasch-CDM and other statistical approaches in terms of assessing students' LP and other cognitive developments.

2 Learning Progressions and Rasch Model

LPs hold many promises to inform curriculum development and instruction and have attracted tremendous attention from researchers in science education. LPs are usually developed based on "big ideas" in science, such as matter, force and interaction, and energy, and highlight students' use of content knowledge in scientific practices (e.g., modeling, argumentation). Though the field has not developed a consensus on LPs, most researchers regard LPs as a tool that depicts students' cognitive development when they learn big ideas. LPs are cognitive models developed based on theories and are verified using empirical data. Conventionally, a hypothesized LP acts as a starting point; empirical assessment data are then collected to validate and refine the assumed LP via psychometric analyses. An LP can be considered as a set of discrete levels or a continuum of cognitive development, both of which are powerful in developing cognitive models. As argued by Jin et al. (2019), neither of them alone provides an all-inclusive description of science learning. LPs are different from construct maps in that they are considered to be more complex (Wilson, 2009). Construct maps specify content being assessed so that they can be relevant to the curriculum (Wilson, 2009). LPs consist of "a set of construct maps, each comprising a 'dimension' of the learning progression," though the complexity of LPs will vary depending on the purpose of the assessment (Wilson, 2009, p. 723).

Researchers have developed different approaches to validating LPs, among which Rasch modeling has been one of the most popular ones (Liu, 2010, p. 212). The Rasch model assumes that there is a linear hierarchy in student performances of a construct.

Constructs denote the latent traits or attributes of examinees that tests intend to assess, such as students' conceptual understanding of buoyancy (Cronbach & Meehl, 1955; Zhai et al., 2021). Instruments and measurement models are needed to assess constructs. Rasch model provides information regarding a person's ability and item difficulty (Johnson & Tymms, 2011; Morell et al., 2017). Furthermore, Rasch modeling transforms raw data into a scale that places "item locations and individual ability estimates on one scale using equal intervals of relative difference" (Osborne et al., 2018, p. 830), allowing for data from different tests on the same metric using common items (Neumann et al., 2013). The Rasch model can be used to plot the distribution of person ability and item difficulty on a Wright map on the same scale, with lower item difficulty and ability towards the bottom and higher item difficulty and ability towards the top (Boone, 2016; Neumann et al., 2013; Herrmann-Abell & DeBoer, 2018). In order for students to correctly answer more difficult items, the logit value of the person's ability and location on the map must be higher than the item difficulty. If the item's logit value is greater than the person's ability, the likelihood for this person to answer the question correctly would be less than 50%. In contrast, if the item's logit value is smaller than the person's ability, the likelihood of answering the question correctly would be greater than 50%. If an item's logit value is equal to the person's ability, then the chances of answering the question correctly would be 50% (Boone, 2016; Morell et al., 2017; Neumann et al., 2013).

Although Rasch model is applied to validate LPs varying on topics, grade levels, and theoretical foundations, the general procedures are identical. LP research usually starts by identifying hypothesized levels. Wilson (2009) termed such levels as a construct map, which includes the highest level, the lowest level, and the messy middle. Generally, researchers rely on one or a combination of three guidelines to identify the hypothesized progression levels:

Curriculum standards as guidelines. Curriculum Standards are usually developed by scientists, education researchers, and expert science teachers, representing a consensus on science learning expectations. Therefore, some researchers analyzed the performance expectations of Curriculum Standards to identify the hypothesized levels of an LP, especially for the upper anchors that correspond to the performance expectations in the Standards. For example, Kaldaras et al. (2021) employed the performance expectation of the Next Generation Science Standards (NGSS; NRC, 2012) to identify the upper anchor of the LP of electrical interactions for high school students. Zhai et al. (2018) employed the Physics Curriculum Standards in China to revise a LP of energy validated by Wei (2014) by removing the two highest levels of the LP in order to adapt the LP to a high school unit.

Empirical findings as guidelines. The most challenging part of an LP is to identify the middle levels that students progress through before reaching the highest level of expectation. Gotwals and Songer (2010) suggest, "while it may be relatively clear to illustrate an endpoint of what we want students to be able to do (e.g., create a scientific explanation about disturbances to a food web), and it is clear when students do not have any knowledge of the content, defining the pathways or middle knowledge that students may have as they move from a beginning point to the endpoint is much trickier given that the middle is often messy and may not be the same for all students" (p. 277). They termed this middle knowledge as a "messy middle," which is most challenging to identify

and assess. In a LP study, researchers can use empirical findings to help organize the “messy middle” to develop the middle levels of a hypothesized LP. For example, Alonzo and Steedle (2009) reviewed literature about students’ misunderstanding of force and motion and analyzed the relations of these misconceptions to develop the middle and lower levels of a hypothesized LP of force and motion.

Learning theories as guidelines. LPs are derived from the research of developmental science, which suggests that learning progresses as students engage in various learning activities. Learning theories are used to explain students’ cognitive development (Lehrer, R., Schauble, L. 2012; Zhai Li, 2021). Some researchers thus explicitly refer to learning theories to help identify the hypothesized levels. For example, Neumann et al. (2013) referred to the theory of the complexity of knowledge (Kauertz & Fischer, 2006) to identify four levels of conceptual understanding: facts (i.e., a fragmental knowledge base indicating that students have pieces of knowledge unconnected to each other), mappings (i.e., pieces of knowledge are simply connected), relations (i.e., qualified connections exist between pieces of knowledge), and concepts (i.e., knowledge pieces are connected, structured, and intertwined). Based on the theory, they identified four levels of energy understanding for their LP: the types of energy, transformations of energy, energy degradation, and energy conservation. Likewise, Osborne et al. (2018) developed their hypothesized LP levels of argumentation based on cognitive load theory (Chandler & Sweller, 1991). They assume that argumentation would demand varied working memory based on the complexity of arguments because the working memory is accountable for students’ LP of argumentation. They thus developed three levels that increase depending on the number of connections students made with claims, evidence, and warrants in their arguments.

Rasch model. Once the hypothesized LP levels are identified, researchers can develop assessment items linked to the levels and then employ the Rasch model to validate the hypothesized LP. In most cases, an LP is centered on one big idea or scientific practice, thus researchers usually assume that the latent trait (i.e., construct) is unidimensional. Thus, a unidimensional Rasch model may be employed to analyze students’ performance on the items. For example, Yao et al. (2017) developed an instrument of Energy Concept Progression Assessment to validate and revise a four-level LP with two variables: key ideas and conceptual development. They found the unidimensional Rasch model best fit their data and assumption. Unidimensional Rasch model allows for the measurement of a single trait (Planinic et al., 2019). Moreover, unidimensional Rasch model involves fewer parameters and thus demands a smaller sample size.

In some instances, however, unidimensional Rasch is insufficient to examine the complexity of the assessment construct; researchers thus have to employ the multidimensional Rasch model. A multidimensional Rasch model assumes that the assessment construct includes subdimensions. That is, an assessment item may address one or multiple subdimensions of the construct. The Multidimensional Rasch model is appropriate for assessments with robust theoretical foundations that support the dimensionality assumption. For example, Hadenfeldt et al. (2016) used a multidimensional Rasch to examine grades 6 to 13 students’ LP on four big ideas of matter (i.e., structure and composition, physical properties and change, chemical properties reactions, and conservation). They compared a four-dimensional Rasch model with two-dimensional and unidimensional Rasch models

and found the four-dimensional Rasch model yielded the most robust psychometric properties, which supported the assumption that the four big ideas are distinct. An in-depth investigation through the four-dimensional Rasch model suggests that students' understanding of each of the four big ideas showed five progression levels, though three of them developed in parallel. Similar research can be referred to by Morell et al. (2017) in which the authors investigated students' LP of the structure of matter using multidimensional analysis. Though multidimensional Rasch may better reflect the authentic construct in assessment, it involves more parameters and requires larger samples than unidimensional models. Therefore, researchers need to determine which Rasch model to adopt based on theoretical assumptions and data.

Given the complexity of LPs, other complementary statistical approaches have also been used alongside the Rasch model. For example, latent class analysis (LCA) has been applied together with the Rasch model in validating LPs. LCA can be used to examine the latent class of the samples based on the similarity of students' responses, which is appropriate to identify or validate students at different levels of the LP (e.g., Gao et al., 2020; Gao et al., 2021). Fulmer et al. (2014) used the Rasch model to analyze high school and university students' LP of force and motion. In this analysis, student responses were transformed into ability measures, which provided evidence with regards to the consistency of student performances on the five-level LP. They then applied LCA to examine the latent classes of student performance, which suggested four classes. Although not consistent with the five levels LP, the LCA result demonstrated that students' average ability progressed along with the four classes.

ANOVA is another analysis that is frequently employed to examine the difference in student performance at different levels of LPs (e.g., Gotwals & Songer, 2013; Todd et al., 2016). Lee and Liu (2009) employed ANOVA and Tukey's post hoc to compare data across sixth-, seventh-, and eighth-grade students taking earth, life, and physical science, respectively. The analysis revealed that the mean knowledge integration increased from sixth to eighth grade, with the mean of eight graders being significantly higher. Testa et al. (2019) also used Rasch, ANOVA, Tukey's post hoc, and exploratory factor analysis to revise their hypothesized LP on quantum mechanics. Through the analyses, they revised their factors to two big ideas, atomic description and measurement and wave function, and its properties in the measurement process.

The development of an LP is usually an iterative process during which researchers revise their hypothesized LP using empirical evidence. In the validation process, measurement models such as the Rasch model play a significant role in linking the observation with the construct. However, Rasch model has limitations, so other methods (e.g., LCA) are usually accompanied. Below, we provide another approach, CDM, that can provide more nuanced evidence to examine students' LP.

3 Learning Progressions and Cognitive Diagnosis Modeling

CDMs are statistical models that provide diagnostic information regarding examinees' mastery patterns of skill sets in problem-solving. Unlike the Rasch model, in which estimating an individual's general ability is the primary goal, CDMs are restricted latent class models that aim to assess an individual's profile on a set of skills used to solve problems (de la Torre & Douglas, 2004). These skills are usually referred to as attributes and assumed to be binary with two possible statuses—mastery and non-mastery. That is, by analyzing students' responses, CDMs provide information about whether the examinee has mastered a given attribute or a combination of several attributes. Individuals' attribute profiles reveal their strengths and weaknesses in problem-solving (i.e., which attributes they have and have not mastered), making them potentially useful for remediation and classroom instruction. For example, by analyzing the buoyancy item referred above, along with other items in the test, CDMs can estimate whether a student is able to (a) identify buoyancy (and gravity), (b) know the direction of buoyancy, and (c) understand that objects stay at rest only if the net force is zero, etc. Students with the same total score or ability value generated through Rasch modeling can have different attribute mastery profiles. Therefore, CDMs could be another complement to Rasch to provide additional nuances of students' cognitive abilities on an LP (Chen et al., 2017; Gao et al., 2020).

The advantages of CDMs in constructing LPs have been recognized in previous studies. For example, Chen et al. (2017) used the rule space model (Tatsuoka, 1983, 1995, 1996) to construct the LP of thermochemistry for senior high school students. Their four-level LP gradually describes how senior high school students master seven attributes of thermochemistry (e.g., exothermic and endothermic reactions, chemical bonds, and heat quantity change). The rule space model is one of the first psychometrical methods for diagnosing cognitive attributes (Gierl, 2007). It is designed to yield specific information about examinees' mastery of specified cognitive skills (Tatsuoka, 1995). With the rule space model, we validated and revised the LP of buoyancy for grade eight students (Gao et al., 2020; Gao et al., 2021). We found that students' mastery statuses of seven attributes featured a four-level LP of buoyancy. For example, students who only grasped the density attribute but failed to master the rest of the attributes were located at the basic LP level. In these studies, students at different LP levels are featured with specific attribute mastery patterns. That is, LP levels are distinguished from each other at the attribute scale.

However, as a special case of the CDMs, the rule space model in analyzing LPs has limitations. First, this model may fail to obtain the correct attribute hierarchical relationships (Yu et al., 2011; Yuan et al., 2022), which consequently affects the validation of LPs. Second, there is no publicly available software or packages to perform the rule space model analysis, which essentially limits its application in practice. To overcome these shortcomings, we put forward the Rasch-CDM approach, which involves both the Rasch model and typical CDM. The Rasch-CDM approach does not utilize hierarchical structures among attributes when fitted to assessment data. Comparisons between attributes' difficulties generated by the Rasch-CDM approach can be used to verify theory-driven attribute hierarchical relationships that correspond to a hypothesized LP. In addition, the

Rasch-CDM analysis can be easily performed using the existing R package. In the following, we present the application of the Rasch-CDM in LPs by taking the buoyancy LP as an example.

4 Rasch-CDM: Integration of Rasch model and CDM for Examining Student Learning Progressions

The Rasch-CDM approach, which was originally proposed in de la Torre and Douglas (2004), includes a two-layer model, where the lower layer is a CDM, which assumes that students' item responses are affected by their attribute mastery profiles (see Figure 2). The upper layer is the Rasch model where the overall ability is assumed to affect students' attribute mastery. A distinguishing feature of this approach is that researchers can estimate students' ability from the Rasch component and students' attribute profiles from the CDM component simultaneously.

The CDM component defines the relationship between attributes measured by an item and the item responses. The CDM we adopted is the generalized deterministic input and noisy gate (G-DINA) model. Technical details of the G-DINA model can be found in de la Torre (2011). To simplify the presentation, we take item 5 from the buoyancy test as an example. Item 5 measures two attributes, A_3 (i.e., A3) and A_6 (i.e., A6). Based on the G-DINA model, the probability of answering item 5 correctly given A_3 and A_6 , each of which can take 0 or 1, is written by

$$P(X_5 = 1 | \alpha_3, \alpha_6) = \delta_0 + \delta_1 \alpha_3 + \delta_2 \alpha_6 + \delta_{12} \alpha_3 \alpha_6,$$

where δ_0 is the baseline, representing the probability of success for students who do not master either required attributes, δ_1 and δ_2 are the increase in probability of success due to the mastery of attributes 3 and 6, respectively, and δ_{12} is the interaction effect due to the mastery of both attributes.

The Rasch component specifies the relationship between overall ability and the mastery of attributes. Let α_{ik} be the mastery status (i.e., 0 stands for non-mastery and 1 for mastery) of

$$P(\alpha_{ik} = 1 | \theta_i) = \frac{1}{1 + \exp [-(\theta_i - b_k)]}$$

where b_k indicates the difficulty of attribute k and θ_i is the overall ability of student i . Note that this is not the traditional Rasch model, which connects the ability with observed item responses; however, we view this as an extension of Rasch model to establish the relationship between attribute mastery and overall ability. It should also be noted that the presented model above is not a multilevel model; rather, it was originally labelled as a higher-order model (de la Torre & Douglas, 2004), similar to the higher-order confirmatory factor analysis.

To validate LPs using this integrated framework, researchers need to define the attributes of interest and then derive hypothesized LPs based on curriculum standards, empirical evidence, and learning theories. The hypothesized LPs may consist of hierarchical relationships among these attributes. A hierarchy between two attributes means that the mastery of one attribute is a prerequisite for the mastery of the other one. The hierarchical relationship can be developed by subject matter experts or by reviewing the literature on the topic. To validate the hypothesized LPs empirically, researchers need to construct test items and develop a Q-matrix that shows the relationship between each item and the attributes. The Q-matrix includes a given number of attributes. To respond to the items, students need at least one of the cognitive skills. The Q-matrix is a binary matrix, with the column indicating the attributes and the row indicating the item. An element 1 indicates an item measures an attribute and element 0 indicates an item does not measure an attribute. After administering the test and collecting student responses, researchers can evaluate whether the data fits the model adequately. In addition to estimating an individual student's ability and attribute profile, the difficulties of all attributes can also be estimated. Based on overall ability, students can be divided into groups with different proficiency levels. Further analysis can be performed to examine the attribute profiles in each proficiency group. Altogether, the attribute mastery proportion of each group, which is derived by calculating the average attribute mastery probability of group students, will be used to refine the hypothesized LPs.

5 A Sample Study: Learning Progression of Buoyancy

In this section, we present a sample study with real data (i.e., the Buoyancy test data from Gao et al., 2020 and Gao et al., 2021) as a practical tutorial for the Rasch-CDM approach. In accordance with the procedure shown in Figure 2, we first present the attributes and a hypothesized LP about buoyancy. We then articulate the item development and Q-matrix construction to assemble a buoyancy test. We employed a data set of 1,127 8th grade students to validate the hypothesized LP of buoyancy.

5.1 Step 1: Identifying Attributes

Buoyancy is an upward force exerted by a fluid or gas due to the pressure difference on different parts of an object, which explains the phenomenon that objects sink and float in liquid or gas (Şahin & Çepni, 2012). Buoyancy is an important concept in everyday life, but it is challenging for students to understand because it includes multiple sub-concepts. Moreover, more than 20 misconceptions may impede students' understanding of buoyancy (Gao et al., 2021). Additionally, advanced reasoning skills are needed for students to use buoyancy to explain phenomena (Inhelder & Piaget, 1958). Thus, it is important for educators and researchers to understand how students progress toward a holistic understanding of buoyancy. The current study employed the Rasch-CDM approach to validate the progression levels from the lower anchor to the upper anchor.

First, attributes relevant to buoyancy were identified. Two important factors were

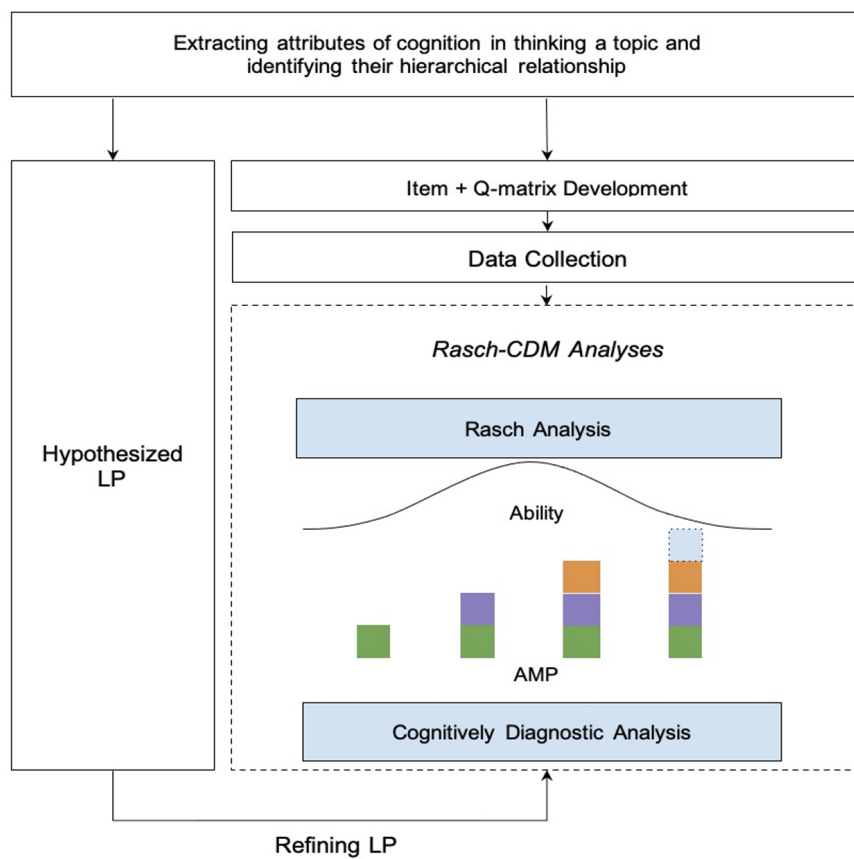


Figure 5.2. Flowchart of using the Rasch-CDM to refine an LP.
Note. AMP: Attribute Mastery Pattern. Q-matrix demonstrates the relationship between the items and the attributes.

recognized when we abstracted attributes of buoyancy: misconceptions and hierarchical property (Gao et al., 2021). Misconceptions help identify critical issues that influence students' understanding of buoyancy whereas hierarchical relationships indicate the premises of understanding the big idea of buoyancy. In Gao et al. (2021), we have presented the rationale and approach to abstracting seven attributes of buoyancy. These attributes and their operational definitions are shown in Table 1. For example, the first attribute Direction indicates the direction of buoyancy is vertically upward regardless of object status (e.g., float or sink). Each of the seven attributes is essential for students to fully explain and predict the buoyancy phenomena.

Table 5.1. The seven attributes of buoyancy and their operational definitions.
Note. Attributes 1 to 7 (e.g., Direction, Identification) are simplified as A1 to A7.

Attribute	Operational Definition
<i>Direction (A1)</i>	Know that buoyancy is vertically upward
<i>Identification (A2)</i>	Can identify not only gravity but also buoyancy that is exerted on an object when it is floating, hanging, or sinking in liquid
<i>Density (A3)</i>	Know that density is a property of an object, which is equal to mass divided by volume and will not change even though the mass or the volume changes
<i>Submerged volume (A4)</i>	Understand the relationship between submerged volume in liquid and object volume
<i>Relation with motion (A5)</i>	Can develop the relationship between the magnitude of buoyancy and the state of motion
<i>Relative density (A6)</i>	Can judge floating and sinking by comparing the densities of the object and of the liquid
<i>Archimedes' theory (A7)</i>	Can employ Archimedes' theory to calculate buoyancy

Alongside the attributes, hierarchical relationships among the seven attributes were also identified according to literature (see Gao et al., 2020 for details). As shown in Figure 3, Direction (A1) and Identification (A2) form the basis of Relation with motion (A5), Density (A3) and Submerged volume (A4) are the basis of Archimedes' theory (A7), and Density (A3) is the basis of Relative density (A6) and Archimedes' theory (A7). That is, students are supposed to master attributes 1 and 2 before they master attribute 5, understand attribute 3 before grasping attribute 6, and know attributes 3 and 4 before they master attribute 7.

After identifying attributes and hierarchical relationships, we developed a hypothesized LP of buoyancy. Two premiers were first identified for constructing hypothesized LP levels, which corresponded to divergent attribute mastery patterns. First, students are more likely to master attribute patterns at lower hierarchical levels rather than those at higher hierarchical levels. For example, students are unlikely to master attribute 5 before knowing attributes 1 and 2 accurately. In other words, the mastery probability of attribute 5 would be enhanced after students acquire attributes 1 and 2. Thus, attribute mastery patterns with a mastery status of attribute 5 and non-mastery statuses of attributes 1 and/or

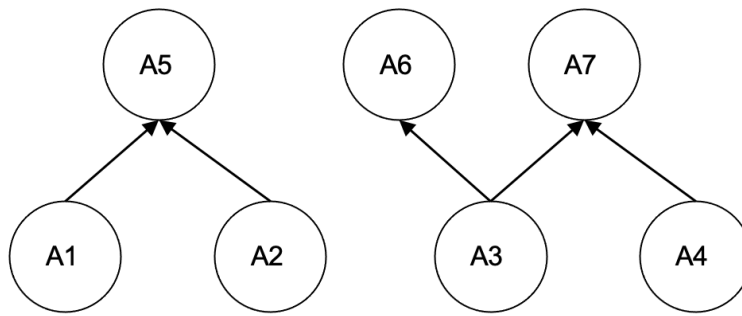


Figure 5.3. The hierarchical relationships of the seven attributes

2 are unexpected in the current study. Second, students at higher LP levels are assumed to display more complex attribute mastery patterns as compared to students at lower LP levels. That is, the higher LP levels are students at, the more attributes they would master.

Based on the two premiers, we adopted a four-level modeling theory to construct our hypothetical LP (Clement, 2008). Science education engages students to use scientific knowledge to construct models to explain phenomena (Zhai, 2022). However, scientific knowledge is not essential for students to construct their models. Students with intuitive knowledge can also develop models to explain phenomena. For example, in Chinese buoyancy indicates “floating” which explicitly suggests an upward force. Therefore, students have no difficulty understanding the direction of buoyancy (A1). We thus assigned A1 to level 1. As students get mature and knowledgeable through learning, they may revise their models using scientific knowledge. In this case, the students first learned density and it would be reasonable for them to use density to explain buoyancy phenomena. We thus assigned A3 (density) and A7 (relative density) to level 2. As more knowledge is accumulated, students would be able to quantitatively explain phenomena. In level 3, the quantitative level, we assume that the students are able to employ Archimedes’ theory to calculate the magnitude of buoyancy. Clement (2008) argued that quantitative modeling was usually more complex than qualitative modeling. Finally, we asserted that the integrative level should indicate the relationship between buoyancy and motion. The topic of force and motion is a big idea that requires students to understand not only buoyancy but also other forces such as gravity and also Newton’s laws. Considering this complexity, the panel split students’ understanding of force and motion into two levels. The attribute A2 is added to level 4. This attribute indicates that students should be able to identify buoyancy by observing the motion of an object. Although this level requires a higher ability, it is qualitative. Whereas in the higher quantitative level—level 5, students should be able to quantitatively establish the relationship between buoyancy and motion.

The hypothesized LP of buoyancy is presented in Table 2, with five levels in total, each of which is associated with a specific attribute mastery pattern. The symbol “X” indicates a mastery status of an attribute; otherwise, the attribute has not yet been mastered at the LP level. For example, students at Level 1 only master attribute 1, and students at Level 2 are expected to master attributes 1 and 2. That is, students at Level 1 only know that buoyancy is a vertically upward force. When students progress to Level 2, they understand that density is a property of an object as well as the direction of buoyancy.

The hypothesized LP of buoyancy indicated that students gradually master more attributes when progressing to higher LP levels (i.e., Levels 3, 4, and 5).

Table 5.2. The attributes corresponding to the hypothesized LP and descriptions

Level	A1	A2	A3	A4	A5	A6	A7	Description
1	X							Students use intuitive knowledge and understand the direction of buoyancy.
2	X		X					Students are able to use density and relative density to explain sinking and floating.
3	X		X	X		X	X	Students are able to use Archimedes' theory to quantitatively explain sinking and floating.
4	X	X	X	X		X	X	Students are able to identify the forces that exert on the objects results in sinking or floating.
5	X	X	X	X	X	X	X	Students are able to use the relations between (net) force and motion to explain sinking and floating.

5.2 Step 2: Item Development and Q-matrix Construction

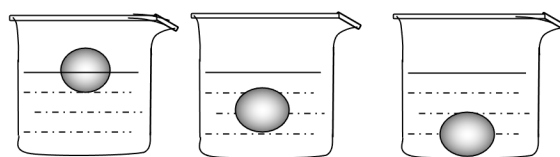
A panel including physics researchers, physics teachers, and measurement experts collaborated to design and revise items to assess students at different LP levels. After conducting a pilot test examining item quality (e.g., item difficulty, item discrimination), one constructed response and thirteen multiple-choice items were included in the final buoyancy test (see Gao et al., 2020 for the test). Table 3 shows attributes measured by each item (i.e., Q-matrix). Most items involved more than one attribute, whereas four items measured only one attribute. For example, item 1 only involved attribute 1, while item 14 measured all the seven attributes. At least four items measure a specific attribute in the buoyancy test.

We used two example items to illustrate item formats and item scoring rubrics. As shown in Figure 4, item 2 is the constructed response item of the buoyancy test, which requires students to identify what forces are exerted on three objects which are floating, hanging, and sinking in liquids (i.e., A2). In contrast, item 14 is one of the multiple-choice items, which measures students' mastery of all the seven attributes. That is, to answer item 14 correctly, students need to understand gravity, buoyancy, and how they relate to objects' motion in fluid (i.e., A1, A2, and A5). In addition, students are able to identify statuses of the two objects in the two liquids (i.e., A3 and A6) when they retrieve densities of water, paraffin, and alcohol from the item stem. Also, students can complete calculations of submerged volumes using Archimedes' theory (i.e., A3, A4, and A7). To score students' answer to item 2, we examined whether they identified both gravity and buoyancy for all the three objects. Students who failed to point out gravity or/and buoyancy for either of the three objects were scored as "0"; otherwise, students were scored as "1". In contrast, the scoring rubric of item 14 is straightforward: students' response was dichotomously scored as "1" for correct and as "0" for incorrect in comparison with the right choice.

Table 5.3. Attributes that correspond to test items

Item	Attributes	Number of Attributes
1	A1	1
2	A2	1
3	A3	1
4	A4	1
5	A3, A6	2
6	A3, A6	2
7	A1, A2, A5	3
8	A1, A2, A5	3
9	A3, A4	2
10	A3, A4, A7	3
11	A1, A2, A3, A5, A6	5
12	A1, A3, A4, A6, A7	5
13	A1, A2, A3, A4, A6, A7	6
14	A1, A2, A3, A4, A5, A6, A7	7

2. Draw the force diagrams for a ball in each of the following states and compare the forces of gravity and buoyancy.



14. Two identical blocks which are made of paraffin are placed into sufficient amounts of water and alcohol separately. What is the ratio of the two blocks' submerged volumes in these liquids? (Note. $\rho_{\text{water}} = 1.0 \times 10^3 \text{ kg/m}^3$, $\rho_{\text{paraffin}} = 0.9 \times 10^3 \text{ kg/m}^3$, $\rho_{\text{alcohol}} = 0.8 \times 10^3 \text{ kg/m}^3$)

A. 9:8

B. 1:1

C. 8:9

D. It cannot be determined

Figure 4. Two example items from the buoyancy test

Figure 5.4. Two example items from the buoyancy test

5.3 Step 3: Data Collection and Data Analysis

We used measurement data from 1,127 8th grade students who took the buoyancy test. These students come from ten middle schools located in five east coast cities in China. All these students used identical physics textbooks, and all had formally learned buoyancy in their science classes by the time the buoyancy test was taken. The test lasted 40 minutes to ensure enough time for students to complete all the test items. Tests in different schools were not conducted at the same time because of a restricted number of test administrators. However, the schools did not communicate about the test, ensuring the validity of the test results. We performed the Rasch-CDM approach using the GDINA package (Ma & de la Torre, 2019) within the R computing environment (R Core Team, 2022). We first examined model-data fit statistics to evaluate whether the Rasch-CDM could fit measurement data adequately. Then, we compared the difficulties of the seven attributes, especially for those within hierarchical structures. Finally, we gathered estimations of students' overall abilities and their attribute mastery proportions generated from the Rasch-CDM analysis. Abilities and attribute mastery proportions provided evidence for the LP of buoyancy, and the hypothesized LP developed in Step 1 was revised and refined. The R codes with annotations have been attached to Appendix A.

5.4 Step 4: Rasch-CDM Results and Refinement of the Hypothesized LP

5.4.1 Model-Data Fit and Q-Matrix Validation

To determine whether the Rasch-CDM fits our measurement data appropriately, we evaluated absolute fit statistics. One criterion of absolute model fit is the standardized root mean square residual (SRMSR; Maydeu-Olivares, 2013). Maydeu-Olivares suggests that SRMSR values smaller than 0.05 indicate well-fitting models. Results showed that the SRMSR value for the Rasch-CDM is 0.044, which is smaller than the benchmark (Maydeu-Olivares, 2013), 0.05. Thus, the absolute fit statistic suggested a robust model fit for the Rasch-CDM.

We also investigated the adequacy of the hypothesized Q-matrix. The Hull method (Nájera et al., 2021) was used to validate the Q-matrix specifications by identifying possible misspecified entries of each item (de la Torre & Chiu, 2016). Results of Q-matrix validation suggest attributes of items 5 and 9 may require modifications. Specifically, item 5 was suggested to measure three attributes (i.e., A3, A4, and A6) rather than A3 and A6, and item 9 was recommended to measure A4 instead of both A3 and A4. Domain experts further examined the two items (see Figure 5) and disagreed with suggested modifications. First, students can answer item 5 correctly without mastering the relationship between submerged volume and object volume (i.e., A4). That is, students can use relative density and the block's status to conclude the block's density. Second, to answer item 9 correctly, students need to understand object density (i.e., A3). Otherwise, students will not be able to compare the volumes of the three blocks, which is critical to compare their submerged volumes. We thus continued to use the original Q-matrix in the following

analyses.

5. Block A sinks in a container with water. As for the density of Block A, which of the following choices is most likely?

A. $0.3 \times 10^3 \text{ kg/m}^3$
 B. $0.5 \times 10^3 \text{ kg/m}^3$
 C. $0.8 \times 10^3 \text{ kg/m}^3$
 D. $1.1 \times 10^3 \text{ kg/m}^3$

9. Blocks 1, 2, and 3 have the same mass, but they are made of different materials: copper, iron, and aluminum (Note. $\rho_{\text{copper}} > \rho_{\text{iron}} > \rho_{\text{aluminum}}$). If we put these blocks into three same containers full of water, they would sink, and the water would flow out. Which of these three circumstances has the most overflow water?

A. Copper
 B. Aluminum
 C. Iron
 D. All are same

Figure 5. Items 5 and 9 which attributes are suggested to be modified.

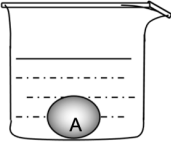


Figure 5.5. Items 5 and 9 which attributes are suggested to be modified

5.4.2 Attribute Difficulty

In addition to model-data fit statistics, the Rasch-CDM analysis estimated the difficulty of each attribute. As shown in Table 4, attribute 3 has the lowest difficulty value (i.e., -1.569), while the difficulty value of attribute 7 is the highest (i.e., 0.024). We also noted that the difficulty of attribute 5 is higher than attributes 1 and 2, attribute 6 is more challenging than attribute 3, and attribute 7 is more difficult in comparison with attributes 3 and 4. These results confirmed the attribute hierarchical relationships established in Step 1. That is, attributes at higher hierarchical levels have higher difficulty values than those at lower hierarchical levels.

Table 5.4. Difficulties of the seven attributes

Attribute	Difficulty
A1	-0.569
A2	-0.238
A3	-1.569
A4	-0.577
A5	0.013
A6	-0.385
A7	0.024

5.4.3 Ability and Attribute Mastery Pattern

To verify the hypothesized LP of buoyancy, we used estimations of overall ability and attribute mastery proportion of the seven attributes, respectively. Given that the total

number of attributes of buoyancy is seven, we assume the highest-level students mastered all seven attributes while the lowest-level students did not master any attribute. Between the lowest and the highest levels, we further segmented students into six groups with equal ability intervals. That is, the range of student abilities was evenly divided into eight groups. For each group, we then calculated students' average attribute mastery proportion per attribute. As shown in Table 5, average attribute mastery proportions were different across the eight groups. The higher average ability one group had, the higher average attribute mastery proportion the group displayed. For example, the average attribute mastery proportions of the first group were all zeros for the seven attributes, whereas the final group mastered all the seven attributes clearly.

Table 5.5. The average abilities and attribute mastery proportions of eight groups

Group	Average Ability	A1	A2	A3	A4	A5	A6	A7
1	-1.390	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	-1.080	0.015	0.029	0.471	0.096	0.081	0.110	0.029
3	-0.744	0.200	0.224	0.744	0.176	0.280	0.168	0.176
4	-0.398	0.390	0.314	0.790	0.448	0.343	0.419	0.276
5	-0.055	0.425	0.485	0.916	0.713	0.335	0.760	0.689
6	0.356	0.832	0.734	0.958	0.832	0.455	0.867	0.713
7	0.626	0.829	0.884	0.973	0.925	0.774	0.890	0.877
8	1.020	1.000	1.000	1.000	1.000	0.996	1.000	1.000

Average attribute mastery proportions shown in Table 5 were used to derive students' attribute mastery patterns. Consistent with our previous studies (Gao et al., 2020; Gao et al., 2021), we employed 0.50 (50% chance to master an attribute) as the cut-off value to determine whether students in a group mastered an attribute or not. That is, if students at a given LP level had an average opportunity that was higher than 50% to acquire a given attribute, the attribute was featured at the specific LP level. According to the cut-off value, the first two groups did not master any attribute; the third and fourth groups only mastered attribute 3; the fifth group mastered three more attributes (i.e., A4, A6, and A7); the sixth group almost mastered all the attributes except attribute 5; the last two groups mastered all the seven attributes. Given that some groups shared identical mastery patterns, we combined them into one group with a unique attribute mastery pattern. We thus merged groups 1 with 2 to form level 1, combined groups 3 with 4 to constitute level 2, and integrated groups 7 and 8 as level 5. Groups 5 and 6 were labeled as levels 3 and 4, separately. Consequently, students were classified into five groups with increasing ability measures and distinct attribute mastery patterns (see Table 6). Generally, students at higher LP levels had higher abilities and were more likely to master the seven attributes. Students at Level 1 did not master any attribute. Average attribute mastery proportions were comparatively low. Students at Level 2 only mastered attribute 3. When students progressed to Level 3, they mastered three more attributes (i.e., A4, A6, and A7) together with attribute 3. Students at Level 4 mastered all attributes except attribute 5. Students at level 4 mastered all the seven attributes at relatively high mastery proportions (i.e., all proportions were larger than 0.90).

Figure 6 shows a map, which indicates students' overall ability distribution (the

Table 5.6. Difficulties of the seven attributes

Level	Average Ability	A1	A2	A3	A4	A5	A6	A7
1	-1.180	0.010	0.020	0.315	0.064	0.054	0.074	0.020
2	-0.586	0.378	0.265	0.765	0.300	0.309	0.283	0.222
3	-0.055	0.425	0.485	0.916	0.713	0.335	0.760	0.689
4	0.356	0.832	0.734	0.958	0.832	0.455	0.867	0.713
5	0.867	0.935	0.956	0.990	0.971	0.911	0.958	0.953

left plot), the attribute difficulty (the middle plot), and the attribute mastery pattern of each LP level (the right plot). We named this map as MGZA (using the authors' Names). The y-axis denotes the student ability and attribute difficulty estimation generated from the Rasch-CDM approach. Using the Rasch-CDM approach, attribute difficulty shared the same scale with ability estimation. As shown in the left plot of the MGZA map, the abilities of most students ranged from -1.5 to 1.0. We noted that abilities calculated from the Rasch-CDM approach had a high correlation with abilities estimated by the unidimensional Rasch model ($r = 0.959$). The middle plot of the MGZA map shows that attribute 3 is the least challenging attribute while attributes 5 and 6 are the most demanding attributes for students. The right plot of MGZA map displays the attribute patterns and the average abilities of the five groups. The size of the dots indicates the probability of students in the group who can master the attribute. Results suggest that much higher LP levels correspond to more advanced attribute mastery patterns.

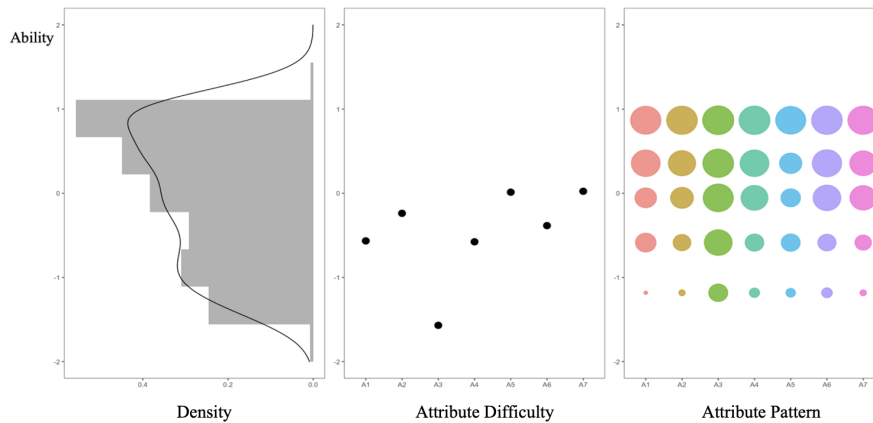


Figure 5.6. MGZA map for student ability, attribute difficulty, and attribute mastery patterns generated from the Rasch-CDM approach

Note. The larger the circle is, the higher proportion an attribute to be mastered.

Attribute mastery patterns inform revisions to the hypothesized LP. As shown in Table 7, the modified LP of buoyancy is different from the hypothesized LP of buoyancy in levels 1, 2, and 3. In conclusion, by combining students' overall ability estimation and their attribute mastery patterns of the seven attributes, this study derived a verified LP of buoyancy.

Table 5.7. Comparisons between hypothesized LP and modified LP of buoyancy

Hypothesized LP of Buoyancy								Modified LP of Buoyancy							
	A1	A2	A3	A4	A5	A6	A7		A1	A2	A3	A4	A5	A6	A7
1	X							1							
2	X		X					2			X				
3	X		X	X		X	X	3			X	X		X	X
4	X	X	X	X		X	X	4	X	X	X	X		X	X
5	X	X	X	X	X	X	X	5	X	X	X	X	X	X	X

6 Discussion

This study presented how to use the Rasch-CDM approach to validate LPs in science education and offers a sample study of buoyancy. Compared with Rasch modeling approach to validating LPs, the proposed Rasch-CDM approach provides more fine-grained information regarding students' cognitive development of thinking scientific concepts. The Rasch-CDM approach measures students' ability with Rasch modeling while simultaneously evaluates students' mastery or non-mastery of different attributes. The combined information can better inform instruction and assessment design because derived LP levels are distinguished from each other at an attribute scale.

In comparison with the rule space model used in previous studies in science education (e.g., Chen et al., 2017; Gao et al., 2020), this study employed the existing GDINA R package (Ma & de la Torre, 2019) that is publicly available to perform the Rasch-CDM analysis. In addition to attribute mastery proportions, the Rasch-CDM approach estimates attribute difficulty which supplied empirical evidence for a theory-driven attribute hierarchical relationship.

However, additional work is needed to further explore the application of the Rasch-CDM approach in LPs. For example, approaches to dividing students into groups affect LP levels. Our previous studies used latent class analysis of students' test data to cluster students into different LP levels (Gao et al., 2020; Gao et al., 2021). Some researchers ordered attributes by the difficulty measure, and attributes with similar difficulties were assigned to the same LP level (e.g., Yuan et al., 2022). The current sample study divided students into eight groups based on the number of attributes. The final LP levels were generated after combining groups that shared the same attribute mastery patterns. The LP levels are slightly inconsistent with findings from prior studies using different methods (i.e., use Rasch modeling and Rule space modeling separately), though with justifiable evidence. In the future, researchers have to select appropriate methods and justify them based on their assumptions and data.

7 Conclusions

The study contributes to the field by introducing a Rasch-CDM approach to validating LPs. This approach extends the Rasch model and incorporates CDMs, which provides

an alternative way to verify hypothesized LPs or refine LPs with additional information. Because the attributes are finer grain-size than assessment items in measuring students' LP, we argue this approach shows advantages over the traditional Rasch approach. The example study provides a practical tutorial to demonstrate the usability of this approach in LP research. The findings in this study support the development of effective instruction and diagnostic assessments to promote students' LP in science education.

Even though this approach provides additional nuance in probing students' LP, we noted one limitation. Different from the Rasch model which uses item difficulty to validate hypothesized LP, this approach starts by estimating students' probability of mastering attributes. Because students' mastery probability may vary by the sample, it is critical that researchers have to recruit a sufficiently large sample size to validate LPs when using the Rasch-CDM approach.

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Author Bio

Authors' information:

Yizhu Gao is a Ph.D. in the Measurement, Evaluation, and Data Science program at the University of Alberta. Her research has focused primarily on assessing the cognitive aspect of learning and problem solving, and developing technology-enhanced learning environments (i.e., adaptive learning systems).

Xiaoming Zhai, Assistant Professor of Science Education and Affiliated Assistant Professor of Artificial Intelligence, Director of the AI4STEM Education Center at the University of Georgia. He is interested in developing innovative assessments and technologies to support science teaching and learning.

Ahra Bae is a graduate student at the University of Georgia in Science Education and a middle school teacher. She is interested in using assessments to support teachers and students in science education.

Wenchao Ma, Assistant Professor of educational research at the University of Alabama. He is interested in item response theory, cognitive diagnostic models and their applications in education, psychology and other social sciences.